

S P E C T R A V I D E O

U S E R S G R O U P -- W g t n

Newsletter Number 18 -- February 1986

The next meeting is on Monday, February 17th at 7.00 in the Staff Common Room, Wellington Clinical School, Mein Street Newtown, the next will be on March 17th.

This month's meeting Darryl's tutorial on data files will take place as it was abandoned in November.

Wellington SV Users Group

Comittee Members

Chairman ----- Darryl Alison Ph 85522 (Paraparaumu)

Secretary ----- Joseph Doi (sorry Joseph)

Treasurer ----- Ross Fletcher 788454

Garry Clark 367872

Newsletter Committee

Steve Hanson 843495

John Matthews 327805

Other specific areas of responsibility for committee members are

Garry -- Radio Access & public domain tape software;

John -- Library;

Darryl -- Program evaluation committee chairman & Kapiti users representative.

Club Subscriptions

The club subscription is now \$15.00/year.

New members this month include Paul Buck, John Landells, and A.E. Fletcher.

Current Membership -- 159.

Note that expiry date for subscriptions is printed on the address label of newsletter. Subscriptions not renewed a month after this date will not receive newsletters from then. Check your label !!

Newsletter Contributions

If you are submitting programs to go in the newsletter they MUST satisfy the following conditions or they will NOT be printed.

1 ... be typed/printed on A4 paper with about 5 lines left blank at top and bottom of page and an inch at least around the sides

2 ... typed with a dark ribbon OR printed using emphasized print mode

3 ... include a description of what the program does , typed on A4 paper with example input

4 ... In the case of programs above 20 lines in length they should be sent on cassette , in this case there is no need for the printout as I can merge them into the newsletter. Enclose a stamped addressed envelope for return of cassette

5 ... line numbers should be continuous -- do a RENUM before printing the programs

6 ... PLEASE MAKE SURE THEY WORK . and would readers please note that I am NOT responsible for programs sent by people which do not work .

7 ... In the case of screen dumps from 3d plotters please indicate the function used and try to get the print as dark as possible -- not always easy I know .

All other newsletter contributions should be sent to The Editor, P.O. Box 26050, Newlands. They may miss the next newsletter if sent anywhere else. Articles should follow 1 and 2 above.

Deadline for Newsletter articles is the 30th/31st of the month previous to publication.

Please follow these guidelines, articles with a 4mm edge all around can not be printed , and neither can articles produced on a dot matrix printer with low resolution . They are simply unreadable after photocopying.

Palmerston North Subgroup

The Palmerston North subgroup are holding meetings on the 1st Monday of each month at THE COMPUTER EXPERIENCE in Palmerston North.

Club Sponsor

MicroStyle Computers -- 29 High Street, Lower Hutt
Action Computers -- P.O Box 7442, Wellington South.

Photocopying from Library

Anybody requiring articles from the library to be photocopied please send request to the LIBRARIAN, not any other committee member. We have had problems with articles not being copied when requested, please ensure that requests go to the librarian.

Also membership renewals should be sent to Box 26050, Newlands, Wellington, not to any other box, otherwise it is possible that the mailing list will not get updated.

SV 738 Owners:

An upgrade of the system disks is now available, including an 80 column BASIC. Contact Action Computers for details.

SV 328 Owners:

A new book, The Magic of SpectraVideo is available now. \$55.00 includes an assembler on tape. Very limited numbers available so be quick.

New Software available now - see advert elsewhere in newsletter.

```
10 :
20 ' BASIC UTILITY PROGRAM FOR MSX AND
30 ' SVI3X8-- CONVERTS BYTES INTO DATA
40 ' STATEMENTS AND BUILDS CODE TO
50 ' LOAD THEM
60 :
61 :
62 ' THIS IS A PUBLIC DOMAIN PROGRAM &
63 ' MAY NOT UNDER ANY CIRCUMSTANCE BE
64 ' REPRODUCED IN COMMERCIAL SOFTWARE
65 :
66 ' DON STANLEY -- ACTION COMPUTERS,
67 ' BOX 7442
68 ' WGTN SOUTH, NZ
69 :
70 INPUT"Enter Beginning Address ";BG
80 INPUT"Enter Final Address ";FL
90 INPUT"Enter File Name ";NF$
100 INPUT"Cassette or disk (C/D) ";DV$
110 X=INSTR("cCdD",DV$):IF X=0 THEN 100
120 IF X<3 THEN OPEN"cas:"+NF$ FOR OUTPUT AS 1 ELSE OPEN NF$ FOR OUTPUT AS 1
130 BH=BG:X=65000!
140 A$=STR$(X)+" "+"DATA "
150 A$=A$+HEX$(PEEK(BG))
160 IF LEN(A$)<220 THEN BG=BG+1:IF BG<= FL THEN A$=A$+",":GOTO150
170 PRINT #1,A$
180 A$="":X=X+1:BG=BG+1:IF BG <= FL THEN140
190 A$=STR$(X)+" restore 65000:for i=&h"+HEX$(BH)+"to &h"+HEX$(FL)+" : reada$:pok
e i,val("+CHR$(34)+"&h"+CHR$(34)+"a$):next:return":PRINT #1,A$
200 CLOSE:END
```

Turbo Pascal vs Basic

Turbo Pascal has been slowly growing in popularity as an alternative to BASIC on the Spectravideo. This is because of some of the advantages it offers over Basic, such as a more logical structure, the ease of writing procedures - user written subroutines which act like Basic's commands, that is variables can be passed to them - and the ease of debugging these features provide. Pascal provides a change of environment from Basic, giving new ideas and often improving Basic programming techniques.

One of Pascal's main weaknesses has always been its lack of graphics and other routines which Basic programmers take for granted. But one of its strengths is the ease with which new routines can be written. Various people in New Zealand are in the process of bringing graphics to the Spectravideo under Turbo's control, Tony Cooper recently released a disc of routines which provided many graphics commands, Don Stanley has recently discovered an easier, and quicker way. By using the same method CP/M uses to access ROM routines, the following procedures provide access to the Line and Paint routines from Basic. This method can be used for many of Basics routines, provided a detailed knowledge of how they work.

This is the beginning of a series on Turbo Pascal programming. Following issues will contain programs written by various club members as well as time and space saving routines, and how they can also be applied to Basic. Contributions from other Turbo programmers would also be appreciated, let us know what you are achieving.

The following program, by Don Stanley, contains routines he has written to access the ROMS. He uses the most useful, and also easiest to access routines, for graphics. The program itself is only a demonstration of these, the routines can be used in any program. While these routines are only as fast as they are in Basic, this is still much faster than compiled user routines, the program itself can show the increased speed of Turbo though.

SELL - OR EXCHANGE

Steve McMert has advised that he has 3 Cartridge Games:
Flipper Slipper, Music Mentor, Frantic Freddy.

If possible, he would like to swap these for other cartridges. If not, they're for sale.

Contact Steve at:- (w) 370109 x8575, or (h) (058)84985.

Because of the Growth in Turbo Pascal one of Action Computers' first offerings is a utility for Turbo programmers. Screen Designer as a utility is different to the usual. It is used to create source code for two purposes.

1. **Show Screen:** By designing a screen from the program, source code is generated to re-create the screen in a program.
2. **Read Screen:** By placing underline characters on the screen, another procedure can be created which inputs variables from the created screen.

Screen Designer takes some of the difficulty out of screen displays and the reading in of values. When the program is finished you are left with a source code file containing a procedure called Show_Screen which displays the screen exactly as it was typed in and, if any variables are to be input, a procedure called Read_Screen which will read those variables in. This file can be read into a program with a simple ^KR from the Turbo editor and compiled with the program.

The procedures generated by Screen Designer are not perfect for every purpose. Everytime the program is run, the procedures generated have the same two names, so if more procedures of the same type are needed at least some of them must be edited - no real problem. The other main problem has a similar solution; Show_Screen is a simple procedure which just uses WriteLn to display any line with text on it from the created screen. If the line does not go to the left of the screen, the blanks there are included in the WriteLn statement.

1e. "WriteIn('This is a partial line ...this too can be edited.');

Read_Screen is the more complex routine and potentially more useful. It calls Show_Screen and then reads each variable from top to bottom of the screen. At the beginning of this procedure is a reminder to VAR the variables it reads in, all other variables are self-contained. It too can be edited if necessary, especially if more than one Show_Screen type routines exist, or if only one variable is to be read in then the arrays used can be changed.

```
program example(input,output);
```

{ TURBO PASCAL PUBLIC DOMAIN UTILITY

This is a demo program which beats writing your own LINE and PAINT routines any day. It switches the ROMS in to operate on user supplied parameters set in the mainline.

Only LINE & PAINT are implemented -- to implement other graphics routines you need to know the entry points and entry parameters. A few have been published in the WGTN SVI Users group Newsletter. Alternatively use the relevant mathematical functions to use LINE to create circles etc.

The program also shows how to manipulate the INTERRUPT routine which is carried out 50 times a second from the ROMS, by using the HOOK at FE79 which the INTERRUPT routine at 3CC2 calls, you can bypass all the usual keyboard checking and so on. This is desirable when doing any graphics from CPM by Bank Switching, as a ^STOP would destroy both the CPM and the temporary BASIC environment, leaving the computer hung !

In the INLINE routines which appear in the PAINT and LINE routines, there is code immediately before calling the switching routines which looks like \$3e/\$00/\$32/\$2b/\$fe , or put another way

```
LD A,0
LD (FE2B),A
```

This is will get rid of any ^STOPS which may be hanging around from before the routine is entered. When you do a ^STOP in CPM this is what happens .

Location FE2B is set to a non-zero value
CPM ignores the ^STOP !

However PAINT (and probably many other routines in ROM) check this location and abort if it is non-zero. so the above routine just makes sure it is zero . This and the interrupt switch off should ensure that you never get lost while the banks are switched

NOTE --- PUBLIC DOMAIN -- These utilities may be used to supplement commercial software if desired and source is quoted. It is not to be sold in its current form.

DON STANLEY / Action Computers / Box 7442 / Wgtn South / NZ }

{-----}

```
type small_byte = 1..15 ;
      string40 = string[40] ;
```

```

var i      : integer;
switching_logic_set_up : boolean ;

{-----}

procedure graphics_screen ;

Var Blank_Bit : byte ;

begin

( set up VDF Graphics 2 mode (SCREEN 1 in Basic). This only sets up the
  registers , use Clear_Vram to initialise things in vram )

  blank_bit := $e0 and $9f ;
  inline($f3/
    $3e/$02/$d3/$81/$3e/$80/$d3/$81/
    $3a/blank_bit/$d3/$81/$3e/$81/$d3/$81/
    $3e/$06/$d3/$81/$3e/$82/$d3/$81/
    $3e/$ff/$d3/$81/$3e/$83/$d3/$81/
    $3e/$03/$d3/$81/$3e/$84/$d3/$81/
    $3e/$36/$d3/$81/$3e/$85/$d3/$81/
    $3e/$07/$d3/$81/$3e/$86/$d3/$81/$fb) ;

end;

{-----}

procedure clear_vram ;

var j : integer;

begin
  inline
  ($f3/$af/$d3/$81/$3e/$40/$d3/$81/      { set up first byte to }
    $01/$ff/$17/$af/$d3/$80/$0b/$3e/$ff/$b8/$c2/*-8/ { write to. Then Fill }
    $af/$06/$03/$d3/$80/$3c/$c2/*-4/$10/$f8/         { pattern tab with 0, }
    $01/$ff/$04/$af/$d3/$80/$0b/$3e/$ff/$b8/$c2/*-8/ { name table as usual, }
    $01/$ff/$17/$3e/$e1/$d3/$80/$0b/$3e/$ff/$b8/$c2/*-$9/ {sprite tables with }
    $01/$ff/$07/$af/$d3/$80/$0b/$3e/$ff/$b8/$c2/*-8/   { 0, color table with }
    $3e/$e0/$d3/$81/$3e/$81/$d3/$81/$fb) ;           { E1 (black foreground }
end;                                                  { and grey background }

{-----}

procedure screen0;      { A La Tony Cooper if you need to set up text mode }
begin                  { when the pictures finished }
  inline($f3);
  mem[$e87f] := $47;
  inline($cd/$6c/$e8/$f3);
  mem[$e87f] := $38;
  inline($fb);
end;

{-----}

Procedure Set_Up_Switching_Logic ;

```

Begin

{ This code is the logic needed to switch the ROMS in and back out when the graphics routine is through . It runs from FFC6 which is unused in the 328 }

```
mem[$ffc6] := $f3 ; mem[$ffc7] := $3e ; mem[$ffc8] := $0f ;
mem[$ffc9] := $d3 ; mem[$ffca] := $88 ; mem[$ffcb] := $db ;
mem[$ffcc] := $90 ; mem[$ffcd] := $f6 ; mem[$ffce] := $02 ;
mem[$ffcf] := $d3 ; mem[$ffd0] := $8c ; { Roms Switch In Routine }
```

```
mem[$ffd1] := $3e ; mem[$ffd2] := $c3 ; mem[$ffd3] := $32 ;
mem[$ffd4] := $79 ; mem[$ffd5] := $fe ; { Interrupt Off} ;
```

{ FFD6 -- FFD8 are CALLS set up by the calling procedure }

```
mem[$ffd9] := $f3 ; { Need this to ensure no interrupts between when
the normal interrupts are restored and CPM is
switched back in }
```

```
mem[$ffda] := $3e ; mem[$ffdb] := $c9 ; mem[$ffdc] := $32 ;
mem[$ffdd] := $79 ; mem[$ffde] := $fe ; { Interrupt On} ;
```

```
mem[$ffdf] := $3e ;
mem[$ffe0] := $0f ; mem[$ffe1] := $d3 ; mem[$ffe2] := $88 ;
mem[$ffe3] := $db ; mem[$ffe4] := $90 ; mem[$ffe5] := $e6 ;
mem[$ffe6] := $fd ; mem[$ffe7] := $d3 ; mem[$ffe8] := $8c ;
mem[$ffe9] := $fb ; mem[$ffea] := $c9 ; { Roms Switch Out Routine }
```

```
mem[$fe7a] := $eb ; mem[$fe7b] := $ff ;
mem[$feeb] := $33 ; mem[$ffec] := $33 ; mem[$ffed] := $f1 ;
mem[$ffee] := $c1 ; mem[$ffef] := $d1 ; mem[$fff0] := $e1 ;
mem[$fff1] := $c9 ; { Interrupt Cut out }
```

{ This last bit of code causes the Basic Interrupt routine to do nothing so that ^STOP and the function keys and other Interruptable things don't destroy the switched environment.

Note -- BASICS 50 times a second interrupt routine starts at 3CC2 by pushing registers, then CALLS a hook at FE79 . The above chunk of code alters that hook by causing a jump to FFEB where the stack pointer is redirected two bytes up to override the RETURN from the hook call, the registers which were PUSHed are POPed, and then a RET returns control to the instruction where the interrupt occurred }

```
switching_logic_set_up := true ;
end ;
```

{-----}

```
Procedur  Draw_A_Line(topleftx,toplefty,botrightx,botrighty:byte;
colour : small_byte ;
style : char);
```

```

var location : integer ;

Begin
  if switching_logic_set_up <> true then set_up_switching_logic ;

  { STYLE determines whether a line/box/filled box }

  if style = 'B' then location := $2452 else
    if style = 'L' then location := $243c else
      if style = 'F' then location := $2401 ;
    mem[$ffd6] := $cd;mem[$ffd7] :=lo(location);mem[$ffd8] :=hi(location);

  { this part sets all the bits that LINE expects to find set up for it }

  mem[$fa3a] := 1 ;          mem[$fa13] := colour ;
  mem[$fe43] := 0 ;          mem[$fe47] := 0 ; mem[$fe42] := botrighx;
  mem[$fe46] := botrighx ; mem[$fe45] := 0 ; mem[$fe49] := 0;
  mem[$fe44] := botrighy ; mem[$fe48] := botrighy;

  { and now run the routine (putting the left (X,Y) pair in BCDE first )

  mem[$0038] := $c9 ; { switch Off CPM's Interrupt facility }

  inline($06/$00/$3a/topleftx/$4f/$16/$00/$3a/toplefty/$5f/
    $3e/$00/$32/$2b/$fe/$cd/$c6/$ff) ;

  mem[$0038] := $c3 ; { Switch CPM's interrupts back on }
end ;

```

```

{-----}

```

```

Procedure Paint(Internx,Interny : Byte ;
  Colour : Small_byte ) ;

```

```

Var I : Integer ;

```

```

Begin
  if switching_logic_set_up <> true then set_up_switching_logic ;

```

```

  { Paint NEEDS

```

```

    ensure FE2B is zero

```

```

  1 .. color in FA13
  2 .. border color in FE41 if in SCREEN 2,color in FE41 in SCREEN 1
  3 .. X pixel in BC and in FE42 and FE46
  4 .. Y pixel in DE and in FE44 and FE48
  5 .. call 2524 to fire it up }

```

```

  mem[$fa13] := colour ; mem[$fe41] := colour ;
  mem[$ffd6] := $cd ;mem[$ffd7] :=$24 ; mem[$ffd8] := $25 ;

```

```

  mem[$0038] := $c9 ; { switch Off CPM's Interrupt facility }
  inline($f3/$af/$47/$57/$3a/internx/$4f/$3a/interny/$5f/

```

```

$ed/$43/$42/$fe/$ed/$43/$46/$fe/$ed/$53/$44/$fe/$ed/$53/
$48/$fe/$3e/$00/$32/$2b/$fe/$cd/$c6/$ff) ;

mem[$0038] := $c3 ; ( Switch CPM's interrupts back on )
end ;

(----- D E M O -----)

begin
switching_logic_set_up := false ;
Graphics_Screen ;
Clear_Vram ;
i := 0 ;
repeat
  Draw_A_line(24,40,128,132,i,'B') ;
  Paint(56,41,i) ;
  i := i + 1 ;
until (i>15) ;
screen0 ;
end.

```

WANTED TO SELL

Spectravideo components for sale; suit person with SV328 wishing to upgrade to disk based system.

2 stand-alone disk drives (155k capacity) \$300 ea.

Disk controller cartridge \$200

Mark I Expander box \$200

Centronics printer interface \$100

80 column video cartridge \$200

Software - Wordstar, Mailmerge, Calcstar, Datastar
Reportstar original manuals and disks \$200

Available separately or the lot for \$1400 o.n.o.

Contact Michael Pollard (proud owner of an SV605B)
Phone Otaki (069) 45194 or write to Box 11817 Manners St
Wellington.

PANDA SOFT INTERNATIONAL Software

We've received a letter from this company who are now marketing SV and MSX software. The list includes a number of games (Esvay - similar to Yahtzee; Seabattle; Chess - with "very good graphics" and a tutor;) and a couple of utility programmes. All available in SV and MSX on tape (\$29.95) and disk (\$34.95) Further details from P.O.Box 4446, Auckland.

No doubt most of our members are now aware that the SV328 and peripherals are available while stocks last at giveaway prices. The notification from the distributors of these "Christmas specials" was received by this Club on 23 December - only just in time for Christmas!

SHORT ITEMS

Apologies to those people who wrote to Deane Whitmore asking for CP/M public domain software and only getting back empty disks. Deane now has the Osborne disks and a list appears elsewhere with details on how to get copies. A few people have been "persuaded" to review some of these disks and write a short note for the Newsletter. What a great way to get copy! The Club has also received three double-sided disks of PD software from the Montreal group which when transferred to single sided disks and we get a source code from SMUG, will be available to members. The letter from CDL who sent them on only said there was an adventure game on one disk but said nothing about the rest. Watch this space!

*

*

*

The Club has recently had a letter from Malcolm McGrane of the Cape Spectravideo Users Group (South Africa) who is keen to make contact with our User groups also to swap newsletters and PD software. He also advises that they are keen to get their own publication going with input from and sales to other countries (like New Zealand). They have 40,000 users in South Africa! I have supplied him with the addresses of our larger groups in New Zealand and also the overseas ones we correspond with.

*

*

*

The bad news is that CDL have advised that they will no longer support the 328 (hence their pre-Christmas sale of 328 hardware at giveaway prices) not that they supported it much anyway - at least in this member's opinion. They will now be pushing the Express - this I've GOT to see! I see this generous move by CDL as making the function of user groups even more important than before.

The good news is that Action Computers who have generously consented to partly support the cost of the newsletter also have a Spectravideo agency and will be able to supply both hardware and software (see December newsletter).

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We've received a letter from Mr. C. Adams of the newly formed Otago Spectravideo Users group who have held their first meeting which a good number of enthusiasts attended. They have the disadvantage in Dunedin that there is no SV dealer to distribute and service SV products. We offered our support and also directed them to the Christchurch group.

SPECTRAVIDEO CP/M PUBLIC DOMAIN LIBRARY FILES

The following is a Disk Directory Listing and copying requirements for the current CP/M PUBLIC DOMAIN files available from the Spectravideo CP/M Library.

Copying is provided free, however, there must be enough included postage with your disks for return mailing.

Single or Double-sided disk formats can be catered for and must be stated with your order. If there is no notice given, then files shall be copied across to a Single-sided format. And only those files that fit will be provided.

Be sure to add the amount of disk space required for each file, as there may be disappointment if some of the files are not provided due to lack of disk space. Allow to send enough disks to cover your requirements for the same reason.

The files listed below have been "squeezed" where possible using SWEEP.COM found on your CP/M SYSTEM DISK. They will have to be "unsqueezed" before execution. The "squeezing" allows far greater disk space available for the storage of data. Any file EXTENSION that has the letter "Q" embedded in it, indicates that the file has been squeezed.

The directories below are of double-sided disks, the files are kept on this format to save disks.

NOTE: Doubled-sided format --> 324k free after formatting
Single-sided format --> 155k free after formatting

A copy of any CP/M Public Domain software can be obtained by writing to:-

Deane Whitmore
3 Lombardy Place
New Plymouth

Provide some form of list of the files you require.

NOTE: If you have any CP/M software to contribute to the Library then send a copy of the files to the above address for inclusion in the Library.

DISK.1.

Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Bytes
AD	CQM	36K	HUNT/A	BQS	2K	OTHELLO	FQR	8K	TOKEN	BAS	2K
ADVENTURDQC		2K	KTAB	DAT	2K	RND115	BAS	2K	TRAVEL	DQT	4K
ADVENTURMQG		30K	LANES	CQM	20K	RTEXT	DAT	2K	TTT	AQM	12K
ATAB	DQT	2K	LTEXT	DAT	2K	STARTREKDQC		4K	TTT	CQM	4K
CALENDARCQM		14K	MAZE	CQM	20K	STEXT	DAT	2K	WUMP	BQS	6K
COMMON	DQT	4K	MSTRMINDBAS		2K	SVMBASICCQM		22K			
EPI/MH	BQS	6K	OTHELLO	CQM	22K	TAXMAN	BAS	4K			

26 Files, occupying 236K of 324 total capacity. 88K bytes free.

DISK.2.

Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Byt
BC	AQM	10K	MX8	COM	4K	QK20	COM	4K	VTERM	DQT	
BC	DQC	2K	MX8NG	COM	4K	QK20	DQC	6K	WSPATCH	COM	
DBL	AQM	8K	MXSET	DQC	4K	QK20	UQD	2K	WSPATCH	DQC	
DBL	DQC	2K	MXSET23	COM	4K	SCRUB3	CQM	10K	WSSPEED	PQT	
DBL-EP	COM	2K	NSWP	CQM	10K	SCRUB3	DQC	18K	WSVIEW	CQ	
DBL-OKI	COM	2K	NSWP	DQC	20K	U-SVDO	CQM	8K	WSVIEW	CQM	
DBL-TESTCOM	2K		NSWP	UQD	2K	VDO	CQM	8K	WSVIEW	DQC	
LU3	CQM	18K	PATCHER	COM	4K	VDO1	COM	4K	XMOD90	AQM	5
LU300	INF	2K	PATCHER	DQC	8K	VDO1	DQC	10K	XMOS-1	AQM	
MODEM1	WQS	2K	PSWORD	AQM	4K	VDO25	DQT	14K			
MX10	COM	4K	PSWORD	DQC	2K	VDONEW	CQM	8K			
MX10NG	COM	4K	PSWRD	COM	2K	VINST	CQM	14K			

45 Files, occupying 314K of 324 total capacity. 10K bytes free.

DISK.3.

Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Byt
CODES	HQL	2K	HELPEXP2ZQR	2K		NAME	NQX	2K	PHONEWS	CQD	
DISPLAY2HQL	2K		HELPEXP3FQT	2K		NEWINDEX	CQD	2K	RECIPE	DQF	
EXPLORE	CQD	4K	HELPEXP3ZPR	2K		NS	CQD	4K	RECIPE	NDX	
EXPLORE1CQD	4K		HELPEXP3ZQP	2K		OUTLINE	DQC	2K	REF	NQX	
FATHER	NQX	2K	HELPEXP4FQT	2K		PERSONS	DQC	2K	SAMPLE	DQF	
FIND	CQD	6K	HELPEXP4ZPR	2K		PHONE	CQD	2K	START	CQD	
FIND	DQC	16K	HELPEXP4ZQP	2K		PHONE	DQF	2K	STARTKITCQD		
GEN	DQC	16K	KITCHADDHQL	4K		PHONE	NDX	2K	STORIES	DQF	
GHOST	FQT	2K	KITCHEN	HQL	2K	PHONEADDCQD	2K		STORY	CQD	
GHOST	ZQP	2K	KITCHGETHQL	6K		PHONEALLCQD	2K		STORYREFNQX		
GHOST	ZQR	2K	KITCHINDHQL	2K		PHONEEDTCQD	2K		TESTSW	CQD	
HELPEXP	FQT	2K	KITCHPRNHQL	2K		PHONEGETCQD	4K		UTILITY	CQD	
HELPEXP	ZQP	2K	LIST	FMT	2K	PHONEMALCQD	2K		UTILITY	DQC	
HELPEXP	ZQR	2K	LIST	ZPR	2K	PHONEOUTCMD	2K				
HELPEXP2FQT	2K		LIST	ZQP	2K	PHONEPRNCQD	2K				
HELPEXP2ZQP	2K		MOTHER	NQX	2K	PHONESCRCMD	2K				

61 Files, occupying 182K of 324 total capacity. 142K bytes free.

DISK.4.

Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Byt
AMORT	BAS	2K	EASTER	BAS	2K	LOVE-P	BAS	2K	POKER	BQS	
AMORTIZRBAS	2K		EPI/MH	BQS	6K	LUNAR	BAS	2K	PROPERTYBQS		
ANOV1	BQS	4K	ETCH	BAS	4K	MADLIBS	BQS	4K	QUEST	BQS	
AVG	BAS	2K	FKDEF	BAS	4K	MATHDICEBAS	2K		RETURN	BQS	
BIO	BAS	4K	GRNFRX	BQS	6K	MONTHS	BAS	4K	REVERSE	BAS	
BLKJK	BAS	6K	HEXAPAWN	BQS	6K	MORTCOMP	BQS	4K	ROCKET	BAS	
CASTLE	BQS	20K	HORSE	BQS	6K	MUGUMP	BAS	2K	ROCKSP	BAS	
CASTLE	DQC	10K	HURKLE	BAS	2K	NAME	BAS	2K	SAP36	COM	
CHASE	BAS	4K	KINEMA	BAS	2K	NICOMA	BAS	2K	STINKFUNDBAS		
CHECKERSBAS	4K		KING	BQS	8K	NIM	BAS	4K	STARTREKBQS		
CHISQ	BQS	6K	KRUSWAL	BQS	4K	NUMBER	BAS	2K	STARTREKDQC		
DATES	BAS	2K	LABELS4	BQS	4K	ONECHECKBAS	4K		TOWERS	BAS	
DEPRE	BQS	4K	LETTER	BAS	2K	ORBIT	BQS	4K	TRADE	BQS	
DEPREC	BAS	2K	LIFE2	BAS	2K	OSCHKBK1BQS	6K		TRAP	BAS	
DISASSV2BQS	6K		LITQUIZ	BAS	2K	PIZZA	BAS	4K	WACC	BQS	
DISKCH	BQS	2K	LOVE	BAS	2K	POETRY	BAS	2K	WORD	BAS	

64 Files, occupying 272K of 324 total capacity. 52K bytes free.

DISK.5.

Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Bytes
2NDMTG	CQL	2K	CPM2CPM	AQM	8K	OSDDAUTOCOM	2K		SPACEIN	AQM	2K
ALLOC	AQM	2K	CPMUTIL	AQM	6K	OSDDAUTODQC	2K		TAX83-1	CQL	10K
AUTO	AQM	4K	DOCUMENTDOC	2K		OWS11	COM	2K	TAXINFO	DQC	4K
AUTOCFM	AQM	4K	FBAD56	COM	2K	PRINTRDYDQC	2K		UNERA15	AQM	8K
AUTORUN	COM	2K	FILECOPYCOM	2K		SAP	AQM	4K	UNERA15	COM	2K
AUTOST	AQM	2K	GROWTH	CQL	14K	SAP36	AQM	8K	UNERA15	DQC	2K
BANNER	AQM	4K	GROWTH	DQC	6K	SAP36	COM	2K	UTL#051	DQC	2K
CHEK	COM	4K	LOANPMT	CQL	4K	SCH-A	CQL	4K	XFER	AQM	6K
CKSUM	AQM	2K	LOCATE	AQM	8K	SCH-B	CQL	4K	XM90DD	COM	6K
CLOSE	AQM	4K	LOCATE	DQC	6K	SCH-E	CQL	8K	XM90SD	COM	6K
COMPARE	AQM	4K	MEX	COM	22K	SCH-G	CQL	4K	XMOD90	DQC	4K
COPYDISKAQM	4K		MEX	HQP	34K	SCH-G-WSCQL	14K				
COPYFILEAQM	4K		MODEM	COM	18K	SCRAMBLEAQM	6K				
COUNT	AQM	4K	OPEN	AQM	4K	SCRAMBLEDQC	2K				

53 Files, occupying 298K of 324 total capacity. 26K bytes free.

DISK.6.

Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Bytes
ALFAROM	QQQ	12K	DU-V83	DQC	8K	HELP	HQP	6K	NUMROM	QQQ	14K
ALGOLM	HQP	8K	EBASIC	HQP	8K	IF	COM	2K	PASCAL	HQP	8K
ASM2	HQP	4K	FBAD	COM	2K	IFSKIP	DQC	10K	RECLAIM	COM	10K
C	HQP	12K	FBAD56	DQC	2K	LIST	AQM	8K	SCRUB2	COM	8K
CBASIC	HQP	10K	FIND+	COM	2K	MAC	HQP	6K	SCRUB2	DQC	6K
CBASIC2	HQP	16K	FIND40	COM	2K	MASM	HQP	6K	SKIP	COM	2K
CFM2	HQP	22K	FIND40	DQC	4K	MBASIC	HQP	14K	SUPERSUBCOM	4K	
DISPLAY	COM	2K	FORTTRAN	HQP	6K	MOVCFM	COM	10K	SUPERSUBDQC	10K	
DU-V83	COM	8K	HELP	COM	2K	NSWEEP	COM	8K	Z80DET	QQQ	18K

36 Files, occupying 280K of 324 total capacity. 44K bytes free.

DISK.7.

Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Bytes
ADVENTURCOM	18K		AIM	DQC	4K	CHES	DQC	2K	FCEXPORTCOM	16K	
ADVENTURDOC	2K		CALENDARCQL	4K		D	COM	4K	PCFILE	COM	32K
ADVENTURDQT	34K		CALENDARDOC	2K		MBASIC	COM	22K	PCFILE	DQC	26K
ADVENTURWQK	24K		CALENDARXQT	2K		NSBASIC	COM	12K	PCSORT	COM	10K
AIM	COM	16K	CHES	COM	26K	OBASIC	COM	16K	PI	COM	10K

20 Files, occupying 296K of 324 total capacity. 28K bytes free.

DISK.8.

Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Bytes	Name	Ext	Bytes
%DATE	MMP	2K	%M	MQP	4K	%MEDTE	MQP	4K	MUMPS	COM	36K
%DIR	MMP	2K	%MO	MQP	4K	%MEDTE1	MQP	4K	SETGLOB	COM	2K
%EDIT	MQP	2K	%M1	MQP	4K	%MTC	GBL	2K	SETMUMPSCOM	4K	
%EDITH	MQP	2K	%M10	MQP	4K	%MTC	MMP	2K	SHOW	MMP	2K
%INDEX	MQP	2K	%M2	MQP	4K	%MTCD	MQP	4K	SPELL	MQP	2K
%INDEX1	MMP	2K	%M3	MQP	2K	%MTCH	MQP	4K	SPELL100MQP	4K	
%INDEX2	MQP	2K	%M4	MQP	2K	%NAMES	MQP	4K	SPELL52	MQP	2K
%INDEX3	MQP	2K	%M5	MQP	2K	ERRORS	DQT	2K	TK	MQP	2K
%INDEX4	MQP	2K	%M6	MQP	4K	FORMD	MMP	2K	TKI	MMP	2K
%INDEX5	MQP	2K	%M7	MMP	2K	FORME	MQP	2K	TKR	MQP	2K
%INDEX6	MQP	2K	%M8	MQP	2K	FORMI	MQP	2K	TKS	MQP	2K
%INDEX7	MMP	2K	%M9	MQP	4K	FORMO	MMP	2K	TURNSTILRQU	6K	
%INDEX8	MQP	4K	%MEDT	MQP	4K	GLOBALS	DQT	2K	WPRNT	MMP	2K
%M	GQL	28K	%MEDT1	MQP	4K	LEXICON	MMP	2K			

55 Files, occupying 208K of 324 total capacity. 116K bytes free.

NOTES ON REINSTALLING WORDSTAR

by Ross Fletcher

(with considerable advice from Graeme Dick)

If you are reinstalling Wordstar and patching to your own requirements the opening instructions using one disk drive are fairly obvious. Providing you have WINSTALL; WS.COM and the two overlay files on the same disk you are in business but the instructions when using two drives are slightly ambiguous.

In this case you are more likely to have WS.COM and overlays on your working disk in Drive B and the WINSTALL program on a separate disk which you should have in Drive A. After typing WINSTALL <return> and passing through the opening paraphernalia to choosing the product (WS) the question is asked where the Wordstar files will be located while you run INSTALL. TYPE B:<return>. At the next prompt you will be asked if the file to call is B:WSU.COM. Type B:WS.COM <return>. If this is not done the system will assume that you are installing Wordstar for the first time and will overwrite the old WS.COM file. If you are doing a first time installation you accept B:WSU.COM and type <return>

[At this point you may get a Bdos R/O error appear. This seems to occur if you use the original SVI Wordstar Ver.3.3 disk but doesn't occur if a COPY of the disk is used. I'd like to hear from anyone who knows why] Ignore the cautionary message about overwriting the existing file. At the end of the installation procedure you are given a further chance to save or reject the changes you have made.

While the installation is menu driven remember that if you are installing Wordstar for the first time and therefore using WSU(ninstalled).COM, YOU MUST CHOOSE A TERMINAL! Unless you have one on the list or know for certain which one is the equivalent of yours, choose the Heath/Zenith 919 which seems to work for most of the monitors sold locally. If you fail to make a choice the Wordstar menus will just look a mess on the screen. If you then go into Custom installation of Terminals you can change the name so that when you boot up in Wordstar your terminal name will appear in the space below the copyright message. The same can be done with the printer name.

The Wordstar features are self explanatory and need no further description from me. Assuming you have a reasonably versatile printer, you can have a great time setting up the Wordstar control codes to perform the things your printer will do.

For example, if you require underlining, Wordstar will underline each individual letter separately. To overcome this change the hex codes for ^PA and ^PN which normally change the type from pica to elite, to those for turning continuous underlining off and on or alternatively use two of the User Patches.

Similarly it is possible to use the ^PY which normally changes the ribbon colour, for the above operation as this command allows different codes for the first and second occurrence. This assumes that you don't have a printer that will do a

colour change!

The controls ^PT and ^PV for carriage roll don't appear to work as most modern dot matrix printers print half-line sub and superscripts and require a different command to turn off sub/superscripting. This perhaps shows how far behind Wordstar is. Graeme discovered that if he put the ON hex codes for sub and superscripts on ^PT and ^PV respectively and the OFF hex code (the same for both sub and superscripts) on User patch #4 ^PR and typed H^PT2^PR^PT he obtained H₂O on the printer. I had no success with my printer and used the ribbon colour for subscripts and two user patches for superscripting.

REMEMBER: If you have a specialised use for Wordstar you can always make a special copy for (say) writing scientific documents and install it for all the features you require for that sort of work. By diving into WSMGSGS.OVR with DDT you can then change the Menus to read the command allocations you have installed. Graeme will be writing an article on this operation.

Finally, make your Wordstar copy using COPY2 (with the spin-off patch) If your drives are as noisy as mine the continual accessing of the disk by Wordstar could send you crazy!

80-COLUMN CARD AND CP/M

In Newsletter No 5 Ross Wakelin explained how to use the PAL80.SUB program to modify CP/M disks for use with the PAL80 television system, which is standard in New Zealand. For the sake of your eyes it is very advisable to amend CP/M disks in this way for use with the 80-column card, whose default is the NTSC standard.

To run this procedure takes several minutes and requires that PAL80.SUB, DDT.COM, SUBMIT.COM and XSUB.COM all be on the sysgened disk which is to be modified. Users may not have realized that having done this to one disk, rather than going through the whole procedure again for each disk, it is much quicker and easier to copy the modified sysgen onto each new disk. The procedure is simply this:

1. If you haven't already done so, take a copy of the CP/M 2.22 disk, type SUBMIT PAL80 (as described by Ross Wakelin) and wait until the procedure has finished running. The result is a PAL80-modified disk.
2. Type SYSGEN at the A> prompt. When asked to insert the source disk, leave the modified disk in.
3. When asked for the destination disk, insert a formatted disk that you want to be modified for use with the 80-column card.
4. Repeat 3 as often as required - once for each disk to be modified. It takes only a couple of seconds for each disk to have the modified SYSGEN put onto it.

Donald Rogers

Altering the PRINT Menu

by Graeme Dick

A previous article by Ross Fletcher has described the basic procedure for reinstalling Wordstar for your terminal and printer. The thought occurred to me, when doing the installation, that it should be quite simple to alter the Print Menu to reflect the use that you have made of some of the commands. It was (is) very easy, requiring nothing more than a passing acquaintance with DDT which is to be used on the messages overlay file, WSMSSGS.OVR.

What Print commands do we have at our disposal. The alterable commands can be found in the WINSTALL.com program under Printer Installation. Each is accessed by a letter given in the menu.

Option O: User-defined functions: There are four User Patches, ^PQ (1), ^PW (2), ^PE (3) & ^PR (4), but there is no room on the Print menu to display any information about them and we can forget them for the meantime.

Option I: Ribbon Selection ^PY: Assuming that you do not have an alternative ribbon color facility (note the American spelling), this is available. I have used it for Italics. Wordstar uses ^PY as a toggle command (same command to turn the function on and off), but during installation you enter your printer-specific on/off commands, e.g. for Italics most Epson-like printers use ESC 4 for ON and ESC 5 for OFF.

Option P: Carriage Roll ^PT & ^PV: These are for Superscript and Subscript. Ross mentioned his difficulty in getting them to work, and briefly described a fix that I had found. The problem is that WS assumes the same printer command turns the facility on and off, and this is not so with the dot-matrix printers I have seen. Thus if you follow the installation, funny things will happen when you try to do Super- or Sub-scripts. For Superscripts, for example, putting your printer's correct turn off code after the superscript (using a User Patch) does not work. You will end up with the next supposed superscript being a Subscript! It seems that WS expects a ^PT (or ^PV for subscripts) to follow the superscript, even though this ^PT is not required by the printer, and you must also enter this in addition to the correct OFF code for your printer.

Option Q: Character Pitch ^PA & ^PN: These can also be used if you don't have the character pitch option. I used them to turn Solid underlining on/off, as an alternative to the ^PS which does not underline spaces between words (you may find this unnecessary, as the same thing can be achieved by typing an underscore character instead of the space).

Now to the point of all this. You can alter the Print Menu for ^PY (Other ribbon color), ^PA (Alternate pitch) and ^PN (Standard pitch). This describes the modifications I made - you can easily make your own changes:

1. With DDT.com and WSMSSGS.ovr on your disk, type: DDT WSMSSGS.OVR <cr> and the DDT prompt (-) will be shown. If you want to scan the file use the D (Display) command.

2. To change Other ribbon color:

The code starts at location 1c44. Note: Spaces are important! Starting at 1c44, you will see the following hex code with ASCII equivalents on the right (I have aligned them to make the change easier):

```
O t h e r       r i b b o n       c o l o r
4f 74 68 65 72 20 72 69 62 62 6f 6c 20 63 6f 6c 6f 72
```

becomes

```
I t a l i c s
49 74 61 6c 69 63 73 20 20 20 20 20 20 20 20 20 20 20
```

Use the S (Set) command to alter the code. Thus, S1c44 will show you the hex code at 1c44 - to change this type the new code followed by <Enter>; to leave it unchanged type <Enter>. When there is nothing more to change, type . (period) to get out of the S function. Hex 20 is the code for a space.

3. To change Alternate pitch:

Start at 1b54 (S1b54):

```
A l t e r n a t e       p i t c h
41 6c 74 65 72 6e 61 74 65 20 70 69 74 63 68 20 20 20
```

becomes

```
S o l i d       U / l i n e       O N
53 6f 6c 69 64 20 55 2f 6c 69 6e 65 20 20 4f 4e 20 20
```

4. To change Standard pitch:

Start at 1ba4 (S1ba4):

```
S t a n d a r d       p i t c h
53 74 61 6e 64 61 72 64 20 70 69 74 63 68 20 20 20 20
```

becomes

```
S o l i d       U / l i n e       O F F
53 6f 6c 69 64 20 55 2f 6c 69 6e 65 20 20 4f 46 46 20
```

5. Exit from DDT by typing ^C or g0.

6. Save the changes by typing SAVE 118 WSMSSGS.OVR. This will overwrite the copy of WSMSSGS.OVR on the disk, so you may wish to make a backup copy of the original file before starting.

That's all there is to it! Next time you run WS and display the Print Menu the changes will be there to remind you of your new functions.

Code for Printer customising: (check yours is the same, if you want the same features)

Standard	Change to	hex code	Comments
I Ribbon selection	Italics ON	1b 34	ESC 4
	Italics OFF	1b 35	ESC 5
O User-defined fns	e.g #1 ^PQ	f	Compressed on
	#2 ^PW	12	Compressed off
	#3 ^PE	e	Expanded (one line only)
	#4 ^PR	1b 54 1b 48	Super/Sub off ESC T ESC H
P Carriage Roll up down	Super ^PT	1b 53 0	ESC S 0
	Sub ^PV	1b 53 1	ESC S 1
Q Alt char pitch ^PA	Solid U/line	1b 2d 1	ESC "-" 1
Std char pitch ^PN	Solid U/line	1b 2d 0	ESC "-" 0

Action Computers

SVI/MSX Software Releases & News FEBRUARY 1986

This is the first of our (hopefully) about bi-monthly newsheets for customers. In these sheets we will be announcing new software releases, upgrades etc. We will also be introducing a heading we hope will become unnecessary quickly ... BUGS.

The newsheet will be posted to all our buyers, plus users groups and software writers, free of charge. So here goes Number 1.

NEW RELEASES

ZIRAKS GEM -- this is a graphics adventure game played over up to 20 levels of dungeon. Features include a series of mazes at each dungeon level, and monsters and other nasties. Available for both SVI and MSX, tape or disk versions are available. Prices are \$24.95 for tape versions and \$27.95 for disk versions.

ENHANCE -- Enhance is a set of 6 new BASIC commands to add to the available set of Basic commands for the SV328. The commands are

CMD SWITCH(bank,byte1,byte2,byte3) which allows use of the extra 32k of ram on a 328 PLUS 32k of ram on a 64k ram card to be used as DATA storage. This does not give more space for Basic programs, but rather allows you to store and retrieve an extra 64k of data, which frees some of the space in the user ram which would usually be data.

CMD KILL which clears the keyboard buffer

CMD SET(value) which causes the bottom <value> lines on the screen to be protected

CMD SPRITE(indicator,sprite1,sprite2) which returns in indicator a value which indicates whether sprite1 and sprite2 are involved in a sprite collision

CMD COPY(string variable) returns string variable with all lower case characters converted to uppercase

The sixth instruction involves the use of the PRT key, and allows you to dump the contents of a video ram mapped screen to the printer. This command is not meant for use with the 80 column card and works only in text mode on Ascii characters from 32 to 126.

Prices are \$24.95 for tape, \$27.95 disk. Available for SVI328 only.

DISASS ----- for MSX only. This is designed to give the basic needs to explore the roms at the cheapest possible price. Allows you to disassemble to screen/printer, to search memory for a string of 2 to 15 bytes, to search memory for a 2 to 15 byte long ascii string, poke values into memory and to obtain hex or Ascii dumps of parts

of the roms. Available for MSX only at a cost of \$24.95 on tape, \$27.95 for 5.25" disk and \$34.95 on 3.5" disk.

In addition to these new titles, we are pleased to announce that MSX versions of Sprite Animator, Graph Plot and Graph Presenter are now available.

PRICES --- we did make the proverbial blue in our first price list. To set the record straight, disk prices for 5.25" disk copies of software are all \$3.00 more than the tape cost, unless other wise indicated. This applies to programs under the ROM BASIC operating systems, not CPM or MSX-DOS. For 3.5" disk versions prices are \$10.00 above the tape price. A corrected price list for all our software is enclosed.

Users Talk --- some valuable feedback from our buyers has been received -- thanks. An upgrade to Graph Presenter to allow scatter plots (line graphs with no lines !) will be implemented shortly. We will comment more on that in the next newsheet. For more comments on Graph Presenter see BUGS (cringe).

We have had enquiries about whether we can sysgen disks. Sorry buyers, thats not possible for various reasons. We do suggest that you copy software on to a working disk and put the master away somewhere safe.

Software For Evaluation -- we are happy to evaluate your software and will market any which we consider of sufficient quality. We now sell overseas as well as in New Zealand, and remember, wherever possible we will undertake to convert your programs to MSX also (or from MSX to SVI of course).

BUGS -- we are still sorting out a few teething problems and Graph Presenter seemed to suffer most. Unfortunately we find that SVI versions of GRAPH3 are completely corrupted. We will be sending out a copy of GRAPH3 to all buyers of GRAPH PRESENTER thus far. Buyers from herein will have the correct copy.

Also the screen dump program for Epsom compatibles in the programs which have screen dumps had a one byte error. In the second data statement (see documentation for line #'s except GRAPH PRESENTER, line 3750) change the EE,1E,4C to read EE,1E,4B. This will allow a full page width plot, but it will truncate the last 15 pixels. To get round this, apply the change as above and change the preceding line so that the E5,3E,1 reads E5,3E,0. The plot will be half page but not stretched downwards. These notes do not apply to MSX.

The documentation for Graph Presenter mentions that F2 and F4 are to look at files. F2 for disk 1, F4 is for disk 2.

Payments Would buyers please NOT send bank notes through the mail, money orders or cheques for the correct amount should be sent.